LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 5: Bubble Sort and Efficiency

Module 2 | 45 minutes | Sorting and complexity | Grades 6–8 · ages 12–14

Objective: Complete bubble sort with nested loops, sort in either order, and measure how the number of comparisons grows with the list size.

Success criteria

- I can implement bubble sort with nested loops.
- I can sort ascending or descending.
- I can describe how its work grows.

Key vocabulary

- **bubble sort**: Repeated passes of neighbour swaps.
- **nested loop**: A loop inside a loop.
- **ascending / descending**: Smallest-first / biggest-first.
- **quadratic**: Work grows with the square of the input.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human bubble sort (5 min): six learners sort themselves by birthday using passes.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Full bubble sort: learners follow the steps.
14 min	**Perform**	Code Lab: bubble sort, Code Lab: sort descending
9 min	**Probe**	Quadratic growth, activity
4 min	**Wrap-up**	Exit ticket: Where must “set j to 1” go, and why?

Unplugged warm-up

Human bubble sort (5 min): six learners sort themselves by birthday using passes.

Teaching the idea

Bubble sort = (n−1) passes of adjacent comparisons; O(n²) comparisons. Reset inner index each pass. Direction of comparison determines order. Contrast with O(n log n) sorts conceptually.

Analogy: Bubbles rising in a fizzy drink: big ones rise first.

Live demo: Sort 4, then 8 items; tabulate comparisons (9, 49) and discuss growth.

Questions to ask

Ask	Listen for
Why reset j each pass?	Each pass starts at the beginning.
How do you sort descending?	Swap the comparison.
Why is bubble sort slow on big lists?	About n² comparisons.

Differentiation

Support
Provide the pass procedure; learners add the outer loop.

Stretch
Stop early when a pass makes no swaps (optimised bubble sort).

Common misconceptions

Misconception	How to fix it
One loop sorts everything	Needs repeated passes.
Doubling items doubles the work	It roughly quadruples.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Where must “set j to 1” go, and why?

Homework
Time yourself sorting 5 cards and then 10 cards by bubble passes. Compare.

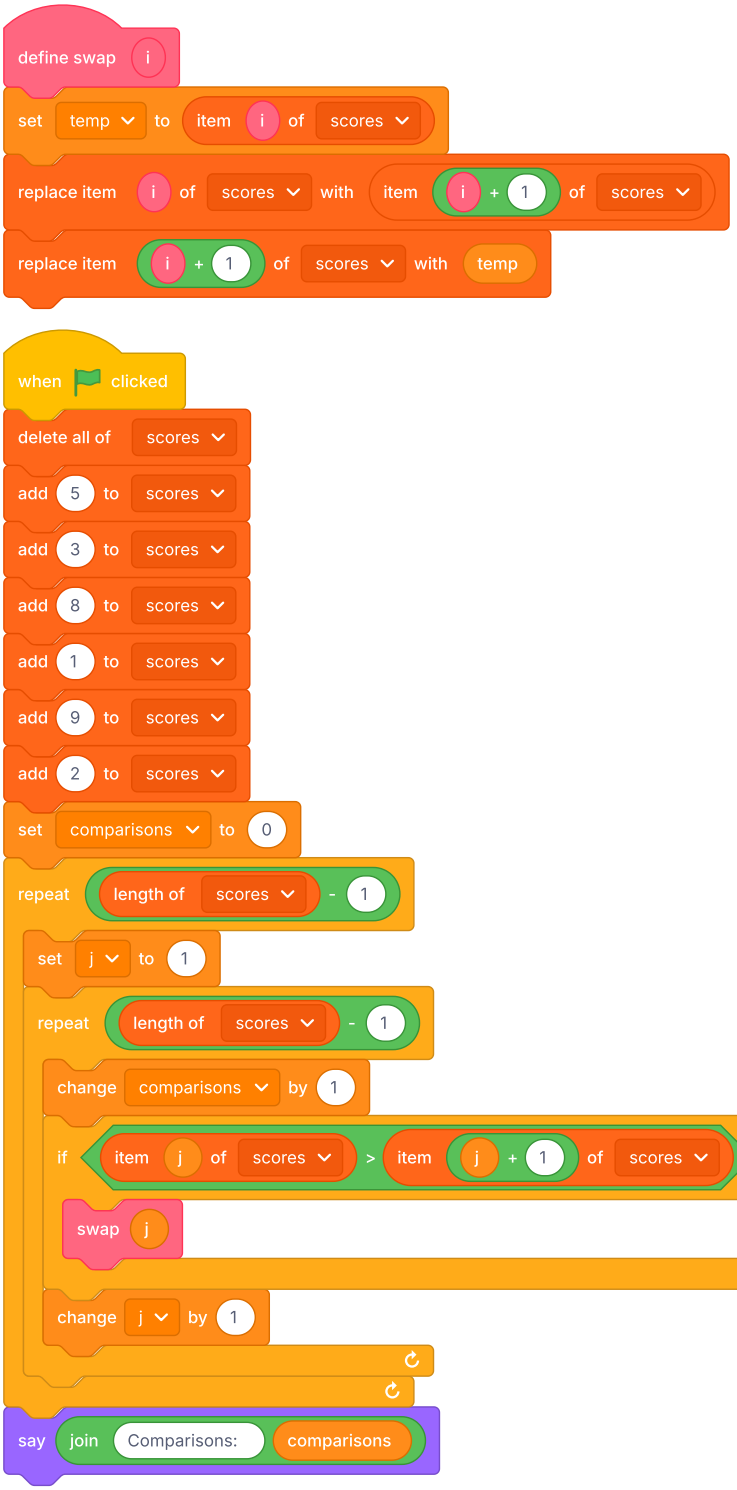
Answer key — Kitabash activities

1. Code Lab: bubble sort (perform · code lab)

Goals checked: Sorted: 1, 2, 3, 5, 8, 9 · Nested loops · 25 comparisons

One solution:

Sorter

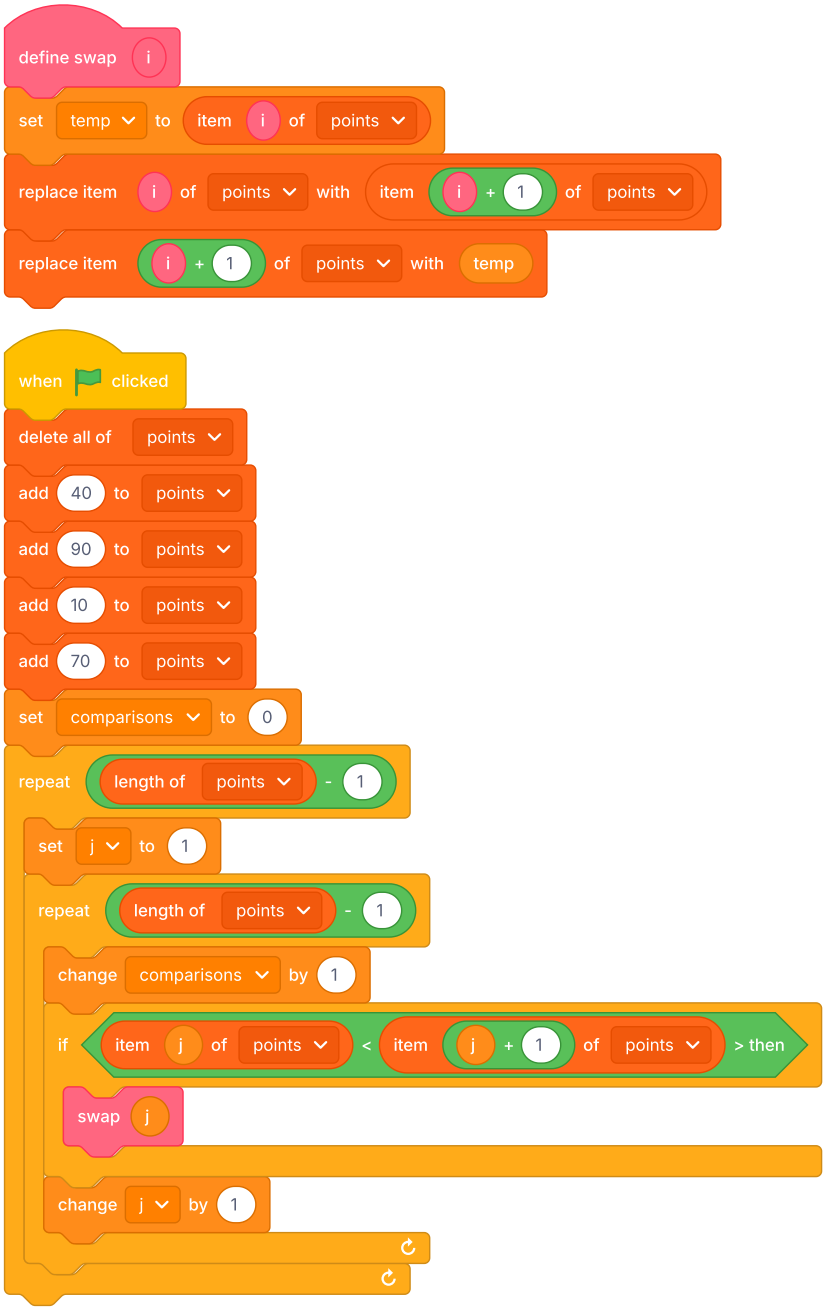


2. Code Lab: sort descending (perform · code lab)

Goals checked: Sorted: 90, 70, 40, 10 · Nested loops

One solution:

Sorter



3. Quadratic growth (probe · multiple choice)

Answer: ~10,000

4. Explain your thinking (probe · free response)

Look for: comparisons, list, slow. **Example:** Bubble sort compares every pair again and again, so the number of comparisons grows with the square of the list size. For a million items that would be far too slow. For a short list, like a class leaderboard of 30 players, it is fast enough and easy to understand.

Answer key — worksheet

1. **Key words:** bubble sort = Repeated passes of neighbour swaps. · nested loop = A loop inside a loop. · ascending / descending = Smallest-first / biggest-first. · quadratic = Work grows with the square of the input.
2. **Plan your code:** Goals: Sorted: 1, 2, 3, 5, 8, 9; Nested loops; 25 comparisons.
3. **Plan your code:** Goals: Sorted: 90, 70, 40, 10; Nested loops.
4. **Quadratic growth:** A. ~10,000 — 10× the items → about 100× the work (quadratic growth).
5. **Explain your thinking:** Bubble sort compares every pair again and again, so the number of comparisons grows with the square of the list size. For a million items that would be far too slow. For a short list, like a class leaderboard of 30 players, it is fast enough and easy to understand.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.